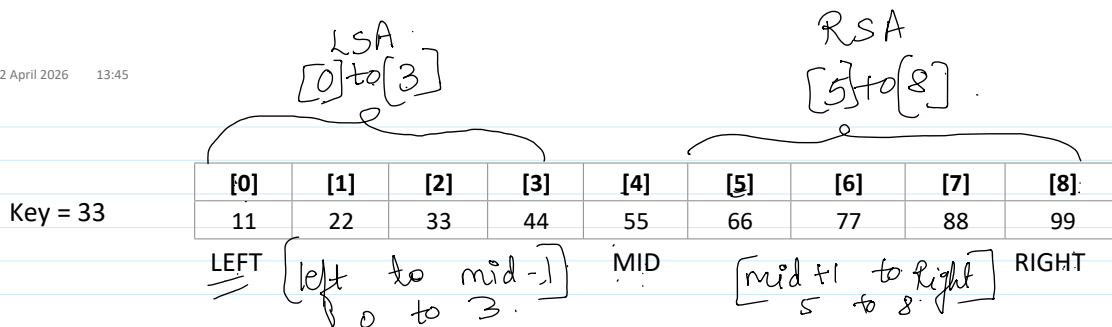




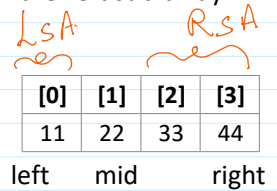
- 1) Get the key from the user.
- 2) Find, the left, right and mid index.
Left : 0
Right : SIZE-1 : 8
Mid : (L+R) / 2 :
(0+8) / 2 : 4
- 3) Compare the key with the mid element.
Key == arr[mid] ?
33 == 55 ? No

If the key matches, return the corresponding index,
Else continue the search.

As the data is in sorted order,
If the key is smaller to the mid element, continue the search in the Left Sub-array.
If the key is greater to the mid element, continue the search in the Right sub-array.
Left Sub-array is considered to be from Left to Mid-1
Right Sub-array is considered to be from Mid+1 to Right.

Repeat steps 2,3 and 4

As the key 33 is smaller to 55, we will continue the search in the Left sub-array



Continue steps 2,3 and 4

- 2) Find left, right and mid.
Left : 0
Right : mid-1 : 3
Mid : (0+3) / 2 : 1
- 3) Compare the key with the mid element .
33 == arr[mid]
33 == 22 ? No
- 4) The key 33 is greater than 22, hence we will continue the search in the right sub-array

[2]	[3]
33	44

Left Right
Mid

Repeat steps 2,3 and 4

- 2) Left : mid+1 : 1+1 : 2
Right : 3
Mid : (2+3) / 2 : 2
- 3) Compare the key with the mid element
33 == arr[mid]
33 == 33
Key found at index 2